

Light Pollution and Sustainability

- Light pollution disrupts the circadian rhythms of organisms
 - Brighter skies from light pollution interfere with migration and breeding patterns of animals (birds, marine life, insects) (McCormack, 2024)
- Light pollution adds to energy costs when outdoor lighting is not shielded, emitting more light than necessary
- Light pollution decreases the effectiveness of ground-based observatories (Varela Perez, 2023)

Light Pollution Research Methods

- Designed walkway wall light covers using cardboard and yellow cellophane
 - Formed the cardboard over lights, made the cover into the same shape, and inserted a yellow light filter into the opening at the bottom of the light cover
- Measured light intensity directly above lights around campus using a Unihedron Sky Quality Meter (SQM)
 - Cloud covering was low, humidity was below 80%, and moon was less than 50% full
- Designed 3D-printed light covers for bollard light poles using Inventor AutoCAD



Figure 1. ERAU wall lights that were used to make the original light covers

Preliminary Results

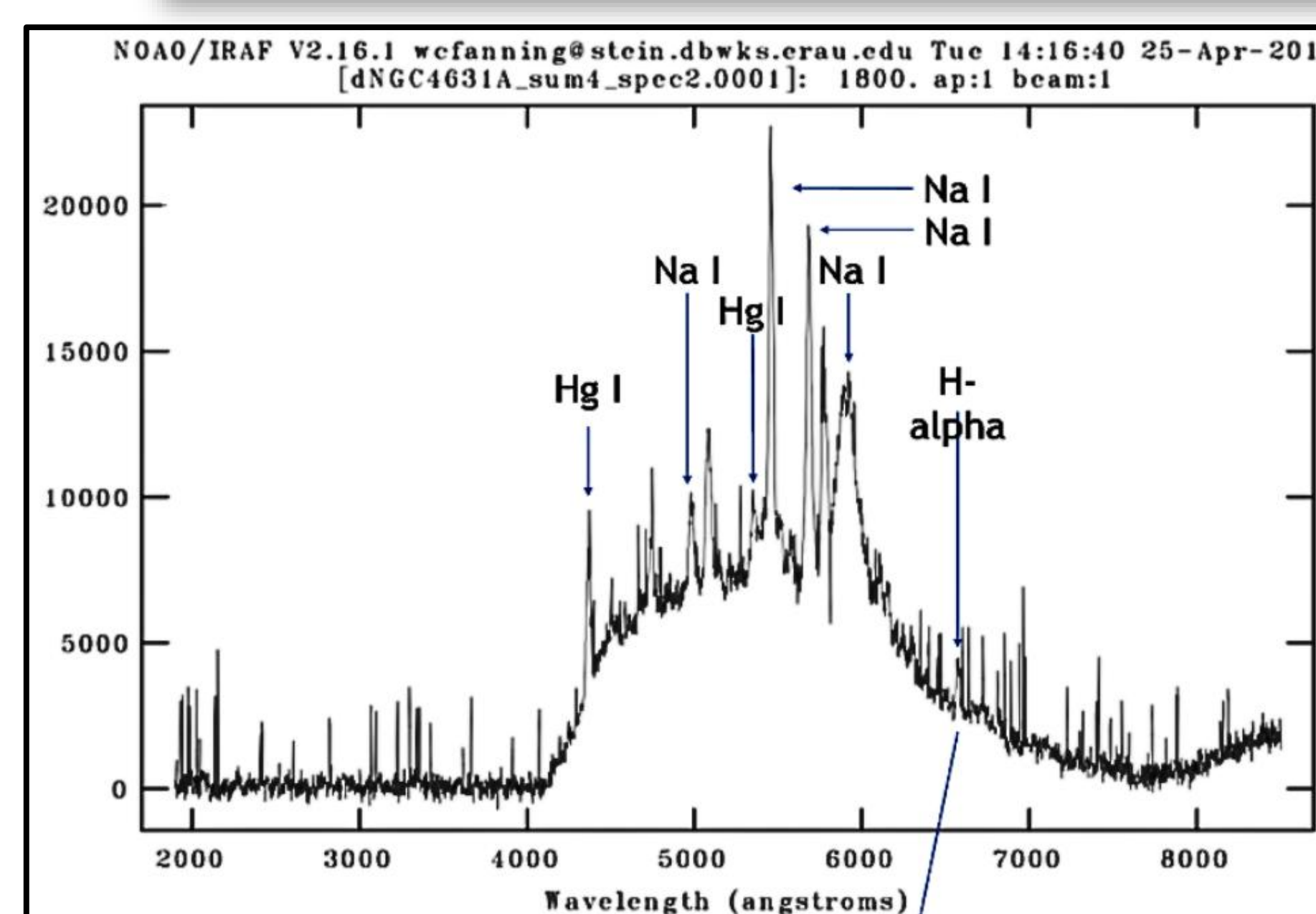


Figure 2.

- Spectroscopic data taken by the 1-meter telescope at ERAU's COAS
- Target signal of a galaxy, labeled as H-alpha, is hard to detect

- Streetlamps contribute to sky glow
- Spectra can be affected by light pollution (seen to left) (Gee et al., 2024)

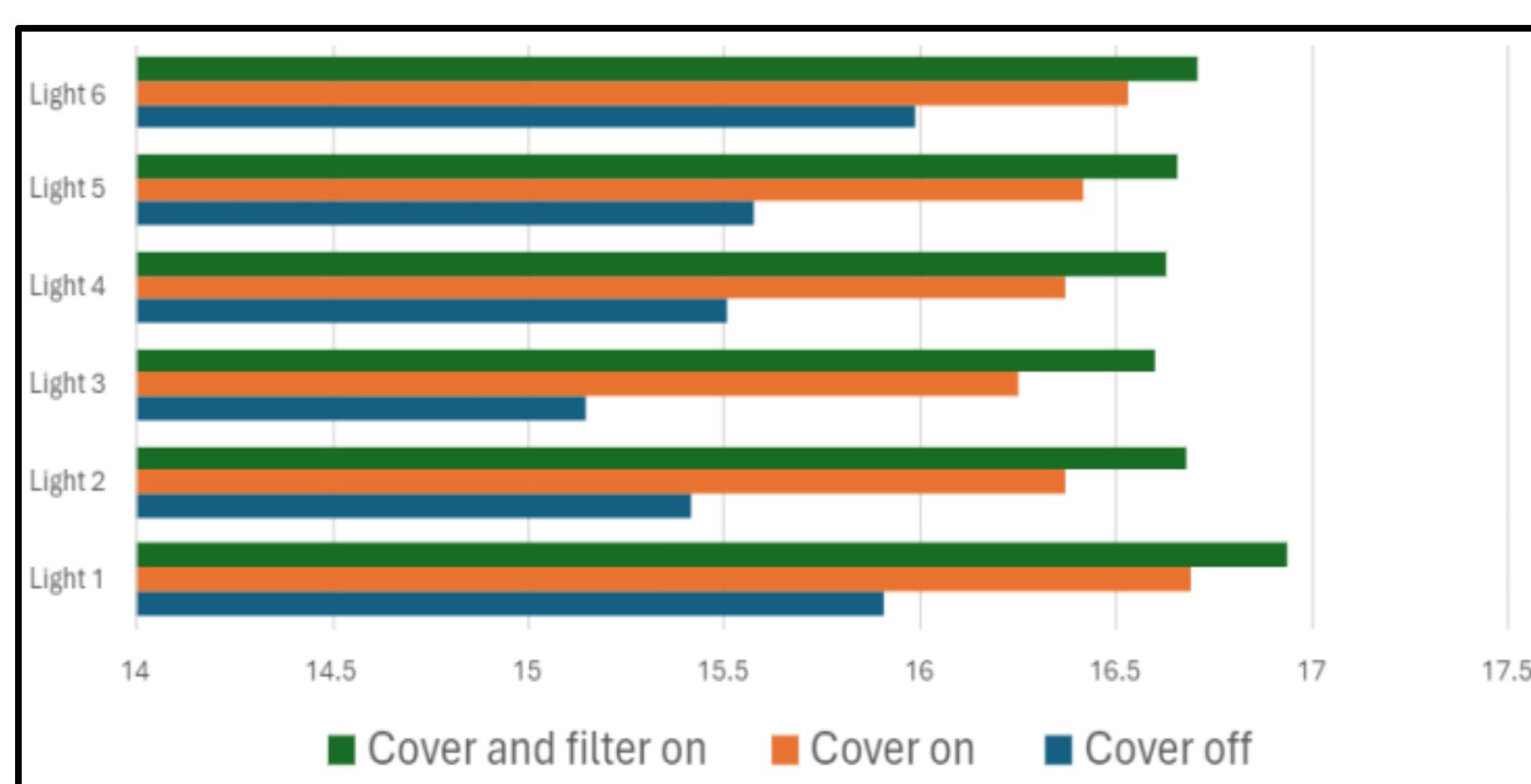


Figure 3. Measurements taken directly above the bollard light poles between College of Arts and Sciences (COAS) and Residence Hall Three (RH3).

- Light covers and filters reduced the amount of light pollution by 1 magnitude (~2.5 times dimmer) (Gee et al., 2024)

References

- Gee, A., Curtis, M., Matney, T., et al. (2024, May 9). *The Impact of Light Pollution on Observational Resources*. Scholarly Commons. https://commons.erau.edu/discovery-day/db-discovery-day-2024/poster-session-2/47/Lamp_Spectrum_and_Light_Pollution_-_Flagstaff_Dark_Skies_Coalition. (2024). Flagstaffdarkskies.org. <https://flagstaffdarkskies.org/critical-dark-sky-issues/lamp-spectrum-light-pollution>
- McCormack, J. (2024, January 29). *Capstone project looks at impact of light pollution | UNC-Chapel Hill*. The University of North Carolina at Chapel Hill. <https://www.unc.edu/posts/2024/01/29/capstone-project-looks-at-impact-of-light-pollution/>
- Varela Perez, A. M. (2023). *The increasing effects of light pollution on professional and amateur astronomy*. *Science*, 380(6650), 1136-1140. <https://doi.org/10.1126/science.adg0269>

Computer-Aided Designs (CADs) for Next Steps

- Measured dimensions of the bollard light poles around ERAU's campus
- Designed CADs in Inventor AutoCAD
- Changed the angle of the shade, the number of covers, and how to connect them
- Converted final design into STL files to prep them for 3D-printing

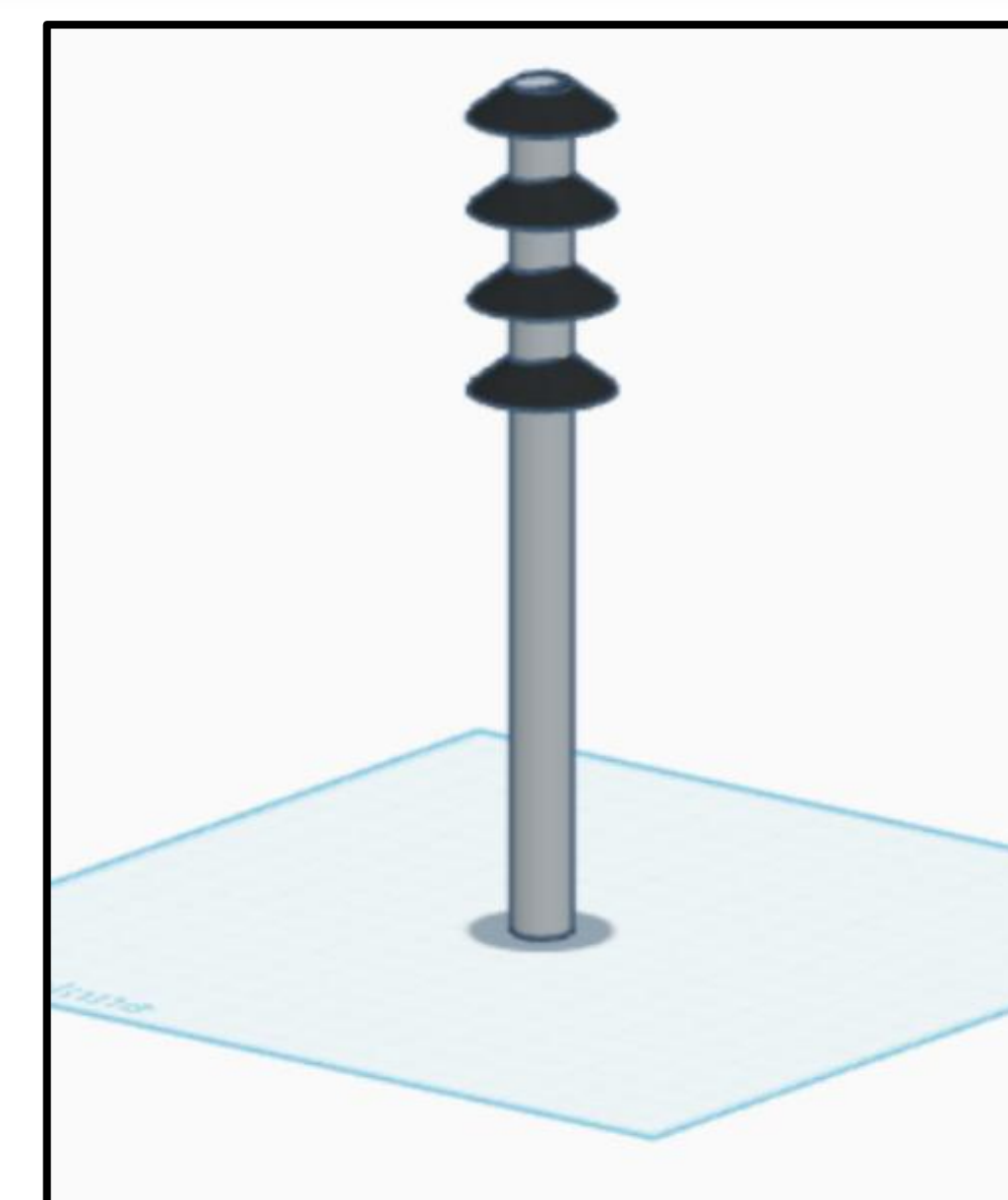


Figure 4. Bollard light pole (found at ERAU) light cover model

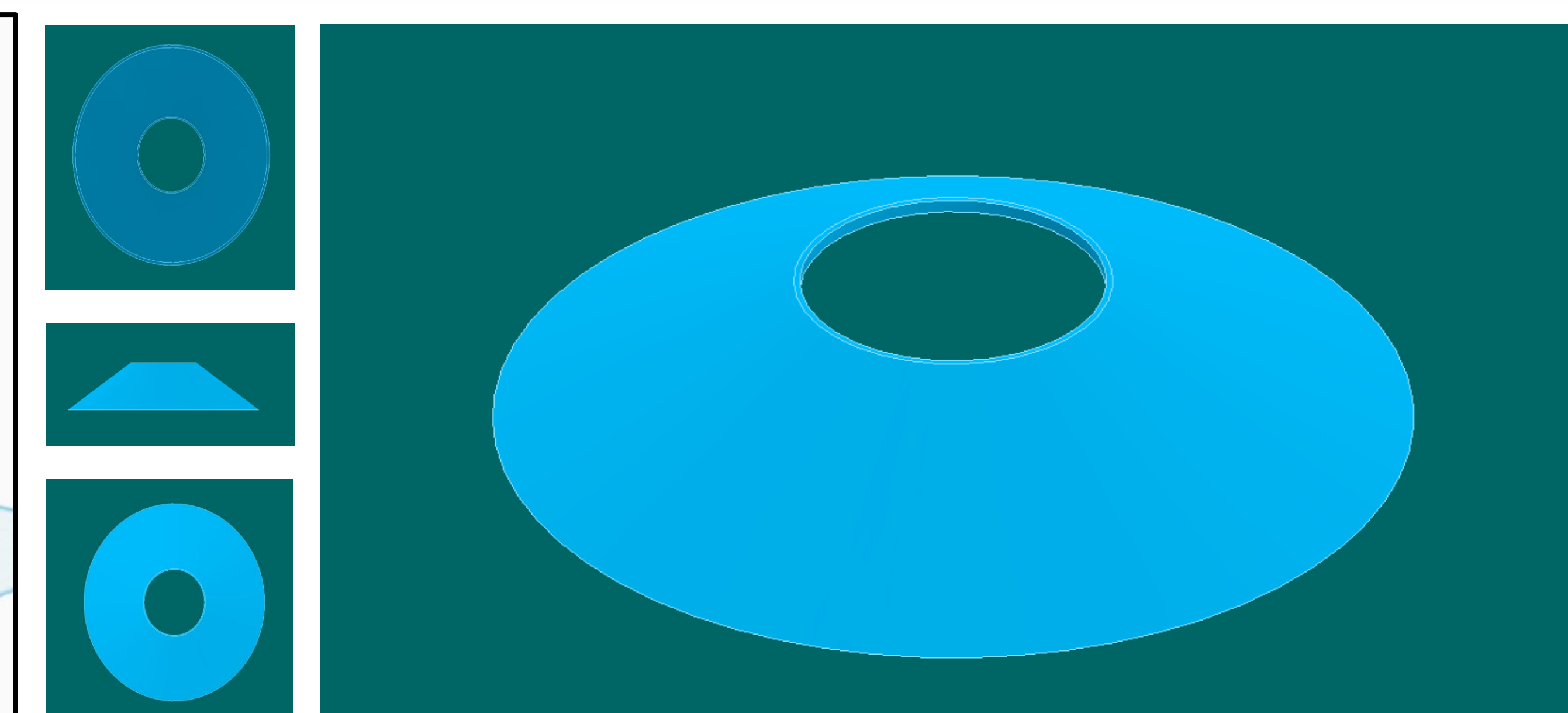


Figure 5. The schematic of the light cover design. Top left image is bottom of light cover, middle left picture is the side, bottom left is the top, and right image is an angled view of the top.

How do Light Filters Reduce Light Pollution?

- White light consists of different colors of optical wavelength ranging from 400-700 nm
- Using a higher wavelength filter (like yellow), only allows light of that wavelength through
 - Higher (redder) wavelengths scatter in our atmosphere less, which reduces skyglow by ½ compared to white lights ("Lamp Spectrum," 2024)
- Light filters reduce the amount of light pointing at the sky

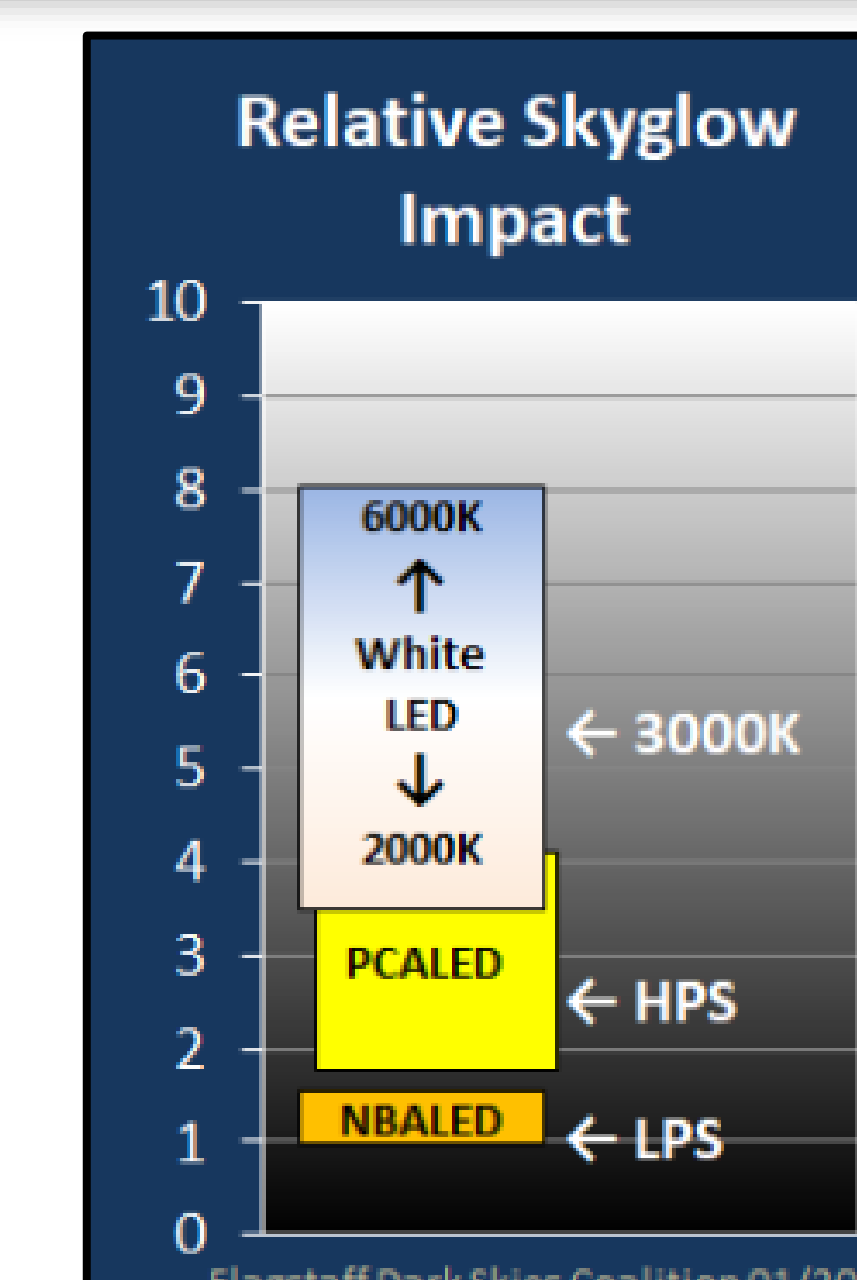
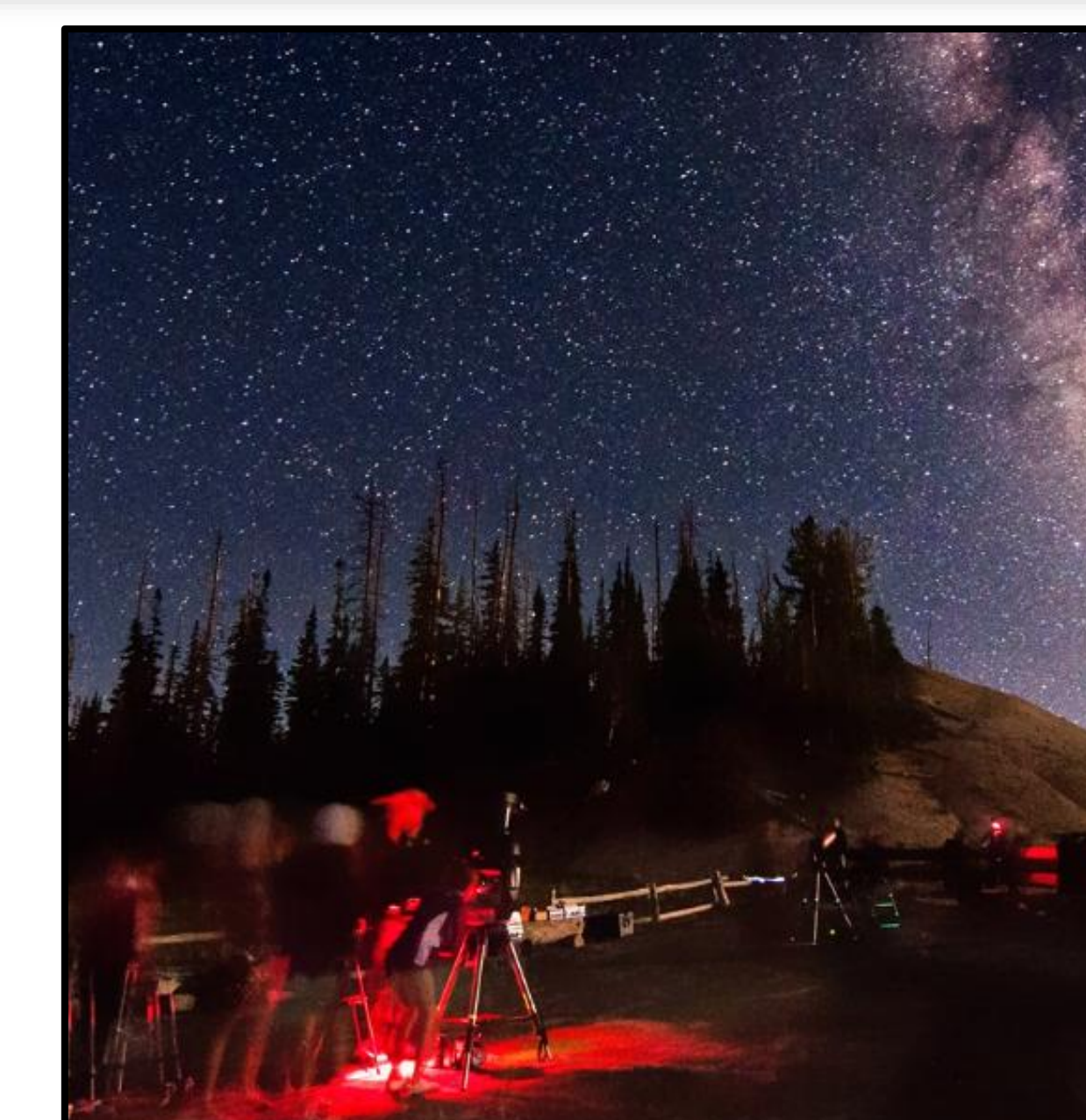


Figure 6. The impact of various types of light sources from Flagstaff Dark Skies, where 0 is no impact and 10 is most impactful. Temperature is inversely proportional to the wavelength, where higher temperatures are bluer light sources ("Lamp Spectrum", 2024)

Future Methods and Goals

- Exploring variations of the 3D-printed light cover designs, adjusting variables such as the angle of the cones of the lamp covers and their diameters.
- Expanding SQM map portion of the project by building and assembling own SQM as a payload for a drone to create a 3D map of measured sky brightness in relation to altitude
- Creating a petition and possibly partnering with ERAU administration to make changes on campus
- Building a case to bring to the City Council of Daytona Beach to enforce current outdoor lighting ordinances

Conclusions

- Natural processes can be altered by brighter nighttime skies and light pollution
- Experimental results at ERAU indicate that there are changes that can be made by implementing mentioned solutions, with the filters and covers reducing the amount of intensity of a light by almost 2.5 times
- There is detailed research on light filters, and their effectiveness has been proven by dark sky reserves
- Making the sky dark again by reducing light pollution will provide astronomers and animals alike with relief

